

# Yonghoon Jeong

Lecturer (Assistant Professor-track), Department of Cyber Science  
Researcher, Institute of Naval Research  
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## Research Interests

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Wireless Communication Systems  
Physical-Layer Wireless Sensing and Networking  
Low-Power Wide-Area Networks (LPWANs)  
Underwater Acoustic Communications and Ocean IoT

## Education

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**M.S. in Electrical and Computer Engineering** 2024 – 2026  
• Seoul National University, Seoul, South Korea  
• Advisor: Prof. Saewoong Bahk  
**B.S. in Electronics and Communications Engineering** 2018 – 2023  
• Kwangwoon University, Seoul, South Korea

## Professional Experience

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**Republic of Korea Naval Academy (ROKNA)**, Changwon, South Korea 2026 – Present  
• Lecturer (Assistant Professor-track), Department of Cyber Science  
• Researcher, Institute of Naval Research  
**Seoul National University (SNU)**, Seoul, South Korea 2024 – 2026  
• Researcher, Network Lab (NETLAB), Institute of New Media and Communications (INMC)  
**Electronics and Telecommunications Research Institute (ETRI)**, Pangyo, Seongnam, South Korea 2023  
• Researcher, AI-SoC Infrastructure Research Section  
**Korea Advanced Institute of Science and Technology (KAIST)**, Daejeon, South Korea 2022  
• Research Intern, Unmanned Systems Research Group (USRG), School of Electrical Engineering (EE)  
• Advisor: Prof. David Hyunchul Shim

## Publications

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### International Papers

- “Receiver-Aware Chirp Waveform Optimization for Mobile Underwater LoRa Communications,” in preparation.
- “On the Symbol Error Rate of Non-Linear Chirp Modulation for LoRa Under Asynchronous Interference,” under review, *IEEE Wireless Communications Letters*. (SCIE, IF: 5.5)
- Yonghoon Jeong, Youngwook Son, and Saewoong Bahk, “On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect,” *IEEE Communications Letters*, vol. 30, pp. 2238–2242, Jun. 2026. (SCIE, IF: 4.5) [Link](#)
- Dongrack Choi, Yubin Choi, Yonghoon Jeong, Jeongyeup Paek, and Saewoong Bahk, “D<sup>2</sup>-PLC: Holistic Design and Implementation of High-Data-Rate Duplex DC Power Line Communication Network,” *IEEE Internet of Things Journal*, vol. 12, pp. 45325–45340, Nov. 2025. (SCIE, IF: 8.9) [Link](#)
- Yonghoon Jeong and Youngseok Choi, “Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos via Diffusion Models,” *Biomedical Engineering Letters*, vol. 15, pp. 575–585, Apr. 2025. (SCIE, IF: 3.4) [Link](#)
- Dongrak Choi, Yonghoon Jeong, Yubin Choi, and Saewoong Bahk, “A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration,” *IEEE ICNP 2025*, Seoul, Korea, Oct. 19–22, 2025. [Link](#)

Domestic Papers

- Yonghoon Jeong and Saewoong Bahk, “A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation,” *The Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Korea, Nov. 20–22, 2024.
- Yonghoon Jeong, “Development of Non-contact Heart Rate Measurement Technology Based on Facial Video,” *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Korea, Jun. 23–25, 2021.

Awards & Scholarships

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|----------------------------------------------------------------|-----------|
| Scholarship Recipient                                          | 2024–2025 |
| • Brain Korea 21 (BK21), National Research Foundation of Korea |           |
| Scholarship Recipient                                          | 2022–2023 |
| • Korea Scholarship Foundation for the Future Leaders (KOSFFL) |           |
| Excellent Paper Awards                                         | 2021      |
| • Korea Computer Conference 2021                               |           |
| University President’s Award                                   | 2020      |
| • Seoul Capstone Design, Seoul                                 |           |
| Scholarship Recipient                                          | 2020      |
| • Samsung Scholarship Foundation                               |           |
| Dean’s List                                                    | 2020      |
| • Highest GPA                                                  |           |

Teaching Experience

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|--------------------------------------------------------------------|------|
| Creative Engineering Design, Prof. Saewoong Bahk                   | 2024 |
| Electronic Circuits Experiment I, Emeritus Prof. Young-chul Chung  | 2023 |
| Electronic Circuits Experiment II, Emeritus Prof. Young-chul Chung | 2023 |
| Electronic Circuits Experiment I, Prof. Youngseok Choi             | 2022 |

Technical Skills

**Programming Languages:** C, C++, Python, MATLAB  
**Frameworks & Libraries:** ROS, ROS2, TensorFlow, PyTorch  
**Simulation Tools:** MATLAB/Simulink, ns-3  
**Development Tools:** Linux, Git, Docker